Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–June 26th, 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Explain the factors that simplify the computation of space complexity in a PRAM application, giving them in order of priority. (1)

B. Please define what is meant by "Work" in a PRAM application. (1)

C. Please describe how **matrix x vector** works when a PRAM architecture is considered. (1)

D. Please describe an example of Common CRCW with a PRAM model. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please list the main differences between the Las Vegas and Monte Carlo algorithms. (2)

B. What are the key properties of the counting sort algorithm? (1)

C. Please list the condition under which the Master Theorem can be used to compute the complexity of an algorithm. (1) **This question can be skipped in case of a passed challenge.**

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the key properties of a Treap and of its search function. (2)

B. Please describe the key properties of the Karger & Stein algorithm. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. What is the Unique Table, and what is it used for in ROBDDs? Does it also have to do with the canonicity of ROBDD? (2)

B. Please describe the "move to front" heuristic (complexity included). Why did we use amortized analysis instead of the worst-case? (2)